THRESHOLD ENTERPRISE SYSTEMS
Production Deployment Guide

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1. CONFIDENTIAL CLASSIFICATION AND OPERATIONAL PURPOSE

*** CONFIDENTIAL RUNBOOK - RESTRICTED TO TECHNICAL ENGINEERING AND OPERATIONS ROLES ***

This document establishes the proprietary production release runbooks, deployment gating checklists, rollback tripwires, and post-deployment validation protocols for Threshold Enterprise Systems. It is classified as CONFIDENTIAL. Access is restricted to authorized roles (ENGINEER, AI_ENGINEER, SECURITY_ENGINEER, ADMIN). Workforce members holding the general EMPLOYEE role are not authorized to retrieve this document within the knowledge base.

The purpose of this Production Deployment Guide is to provide a standardized, rigorous, and repeatable release governance framework that ensures all software updates, cloud infrastructure modifications, microservice deployments, and AI service releases transition safely from staging to production with zero unplanned service disruption, verifiable security compliance, and guaranteed instantaneous rollback capability.

In accordance with corporate governance mandates, this guide strictly omits live production IP addresses, credentials, server hostnames, or cloud account keys, focusing entirely on operational architecture, verification gates, and release governance.

2. Scope and Deployment Archetypes Covered

This runbook applies to all software engineering squads deploying services into production clusters across our multi-region cloud infrastructure:

1. Backend Application Services: Containerized Python microservices, REST APIs, and background worker queues.

2. AI and RAG Inference Pipelines: Vector database clusters, embedding worker nodes, LLM inference endpoints, and guardrail microservices.

3. Database Schema Migrations: Relational database schema updates, index builds, and data migrations governed under ENG-005.

4. Cloud Infrastructure and Mesh Changes: Kubernetes ingress configurations, service mesh routing rules, and cloud infrastructure updates provisioned via Terraform.

3. The Seven Mandatory Pre-Deployment Verification Gates

Before any production deployment pipeline can be authorized, the Release Manager and Squad Lead must certify seven prerequisite gates:

Gate 1 - Code Review and Ownership Sign-Off (ENG-002 Gate): All PRs merged into the release candidate must possess a minimum of two peer approvals and verified Code Owner sign-offs.

Gate 2 - Automated CI/CD Testing Certification: The release build must pass 100% of automated unit tests, contract tests, and end-to-end integration tests in the staging environment with zero open blockers.

Gate 3 - Security Vulnerability Clearance (SEC-001 Gate): Container image scans must certify zero Critical or High CVEs in application dependencies and base operating system layers.

Gate 4 - AI Governance Certification (For AI Services): AI models, RAG pipelines, or LLM wrappers must hold active Model Approval Certifications under AIG-003 and comply with deployment standards under AIG-010.

Gate 5 - Database Migration Validation (ENG-005 Gate): All schema migrations (Alembic) must be tested against an anonymized staging database replica, verifying execution times under 60 seconds and zero table locking.

Gate 6 - Configuration and Secrets Audit: All environment variables, feature flags, and secrets must be verified in HashiCorp Vault with zero hardcoded credentials.

Gate 7 - Documented Rollback and Runbook Plan: A verified, step-by-step rollback runbook must be published in the release ticket with tested automation scripts.

4. Zero-Downtime Database Migration Protocols (Expand-and-Contract)

Schema updates in distributed production environments must never require database downtime or cause table-locking freezes:

- The Expand-and-Contract Architectural Pattern: Schema alterations must be decoupled into two sequential deployments. In Phase 1 ('Expand'), new nullable columns or tables are introduced while existing application code continues operating normally. In Phase 2 ('Contract'), once new code is verified and data backfilled, deprecated legacy columns are safely dropped.

- Non-Locking Index Creation: In PostgreSQL production databases, all index additions must execute using CREATE INDEX CONCURRENTLY to prevent blocking concurrent write transactions.

- Backward-Compatible Migration Scripts: Alembic migration scripts must provide deterministic downgrade() routines tested in staging to guarantee reversibility in the event of an immediate release rollback.

5. Staged Deployment Strategies: Blue/Green and Canary Rollouts

Direct in-place replacement of live production instances is strictly prohibited. Production releases must employ Blue/Green or Canary deployment strategies:

- Blue/Green Deployment Architecture: The active production environment (Blue) continues serving live traffic while the new release (Green) is provisioned in parallel. Synthetic smoke tests validate Green environment health, database connectivity, and API responses. Once validated, load balancers cut over traffic from Blue to Green. Blue remains warm on standby for seventy-two (72) hours to guarantee instantaneous one-click rollback if subtle regression bugs manifest under real production workloads.

- Canary Traffic Rollout Progression: For high-traffic public API endpoints and AI inference services, traffic is shifted incrementally using Istio service mesh virtual services: 5% canary traffic for 30 minutes -> 25% traffic for 30 minutes -> 50% traffic for 30 minutes -> 100% full production cutover.

- Automated Traffic Analysis: During each canary stage, automated observability daemons analyze error rates, P99 latency, and CPU/memory profiles against the baseline cluster. Any statistical deviation triggers automated canary abortion and traffic rollback without human intervention.

- Service Mesh Traffic Splitting: Header-based routing and cookie affinity allow internal engineers and QA specialists to test canary pods live in production before external user traffic is introduced.

6. Automated Rollback Tripwires and Execution Procedures

During deployment and the subsequent soak period, automated monitoring daemons immediately execute automated rollback if any of the following tripwires fire:

- HTTP 5xx Error Spike: Error rate exceeds 1.0% of total request volume over a two-minute window.

- Latency Degradation: 99th percentile (P99) API latency increases by more than 25% over baseline.

- Memory Leak or CrashLoop: Service pods experience OOM-kills or restart loops on canary nodes.

- Database Connection Pool Exhaustion: Database connection wait times exceed 200 milliseconds.

- AI Guardrail or Hallucination Breaches: Automated prompt security monitors (AIG-007) detect guardrail bypass or extreme hallucination spikes.

Rollback Execution Runbook: Upon tripwire trigger, the deployment orchestrator terminates canary pods and redirects 100% of traffic back to the standby Blue cluster within thirty (30) seconds, notifying the engineering on-call channel.

Post-Rollback Triage: When an automated rollback occurs, the release state is immediately frozen. Engineers must conduct a forensic post-mortem in the staging environment before any rescheduled deployment attempt.

7. Release Windows, Deployment Freezes, and CAB Approvals

To minimize user disruption and maintain peak engineering readiness during releases, strict scheduling controls apply:

- Standard Deployment Windows: Standard production releases must be executed Tuesday through Thursday between 10:00 AM and 3:00 PM local regional time (IST for India, PST for US, GMT for UK, SGT for Singapore). Deployments on Fridays, weekends, or outside core hours are strictly prohibited.

- Seasonal Deployment Freezes: Formal deployment freezes occur during global enterprise financial quarter-ends and major commercial holiday periods. Only critical hotfixes authorized by the Emergency Change Advisory Board (ECAB) are permitted during freeze windows.

- Change Advisory Board (CAB) Review: Routine production releases require formal CAB review and sign-off 24 hours prior to deployment, verifying prerequisite checklists and rollback readiness.

8. Post-Deployment Verification and Hypercare Window

Following 100% traffic cutover, the engineering squad must execute formal post-deployment validation:

- Synthetic Smoke Tests: Automated suites execute core business journeys (e.g., user login, document upload, RAG search query, API transaction) to verify functional integrity across global regional endpoints.

- Log Inspection: SRE on-call inspects central log aggregates for unexpected warning clusters, unhandled exceptions, or database query timeouts.

- 2-Hour Hypercare Monitoring Window: The authoring engineers and SRE on-call remain active on the deployment bridge for 120 minutes following cutover, monitoring real-time Grafana dashboards.

- Change Ticket Sign-Off: Once hypercare concludes with zero defects, the release ticket is marked CLOSED in the enterprise change management system.

- Automated Health Reports: Synthetic monitoring daemons generate an automated deployment summary report detailing error distribution, latency distributions, and resource consumption deltas.

9. Emergency Rollback and Incident Escalation

If a defect surfaces post-hypercare, engineers must not attempt live ad-hoc patching in production ('cowboy coding').

Instead, the Incident Commander must initiate the rollback runbook or trigger the Engineering Incident Management Runbook (ENG-006) to investigate and deploy a formal hotfix.

Cross-Document Governance Interconnections:

- ENG-002: Code Review Policy (Enforces peer review completion prior to deployment).
- ENG-005: Database Access Policy (Governs production database migration approvals).
- ENG-006: Engineering Incident Management Runbook (Governs incident response upon deployment failure).
- AIG-003: AI Model Approval Policy (Prerequisite approval gate for AI system releases).
- AIG-010: AI Model Deployment Policy (Defines specialized AI model release governance).